

Progress Report: Smart Veterinary System with AI-Based Disease Detection

Course Code: CSE 299

Section: 04

Project Group: 05

Group Members:

2311677042 – Nafia Haque Muskan

2312559042 – Rishat Bin Sultan

2232239642 – MD Tamim Hasan

2311101642 – Fariha Islam

Abstract—This progress report presents the development progress of the Smart Veterinary System with AI-Based Disease Detection. During this reporting period, the team concentrated on building the shop-related features of the web platform. The completed work includes account management, profile creation and editing, cart management, payment handling, and order history functionality. Both frontend page development and corresponding backend implementation were completed using Django.

I. CURRENT PROGRESS

During this week, the team focused on developing the shop module and related user account functionalities. Each member was responsible for both the frontend design and the backend integration of the assigned features.

TABLE I
WEEKLY PROGRESS AND RESPONSIBILITIES

Completed Task	Responsible Member(s)
Designed and implemented the shop homepage and product display section in frontend and backend	Rishat Bin Sultan
Developed account-related pages and backend handling for user account management	Nafia Haque Muskan
Created profile and edit profile pages with Django backend integration	Nafia Haque Muskan
Implemented add-to-cart functionality with frontend interface and backend logic	Fariha Islam
Developed payment page and integrated backend payment handling structure	MD Tamim Hasan
Implemented order history page and backend logic for tracking previous orders	MD Tamim Hasan
Managed product, cart, and order-related database operations in Django models	Fariha Islam
Performed feature integration, testing, and debugging for shop-related modules	All Members

II. CHALLENGES AND DECISIONS

Challenges:

- Maintaining smooth coordination between frontend pages and backend functionality for the shop system.

- Handling cart updates, payment flow structure, and order tracking in an organized way.
- Ensuring profile and account-related features work properly with user authentication.

Decisions:

- The team decided that each member would handle both the frontend and backend of their assigned module for better consistency.
- Since the frontend development is nearly complete, the next development phase will focus mainly on AI-based disease detection.

III. FUTURE PLAN

In the next phase, the team will:

- Focus on developing the AI-based disease detection module.
- Prepare and preprocess disease-related pet datasets for model training.
- Implement machine learning models using Python and Scikit-learn for disease prediction.
- Integrate the trained AI model with the Django backend for real-time prediction.
- Test and evaluate the prediction performance of the model for accuracy and reliability.

IV. CONCLUSION

This week, the project development mainly focused on the shop system and user account functionalities. Important features such as account management, profile creation and editing, add-to-cart, payment, and order history have been developed successfully. As the frontend work is now almost complete, the next stage of the project will mainly focus on implementing the AI-based disease detection system using Python and Scikit-learn models.